From African-Natural-Product Chemical Space to Resistance-Resilience Hypotheses: An Integrated Computational Study of Antimalarial Polypharmacology

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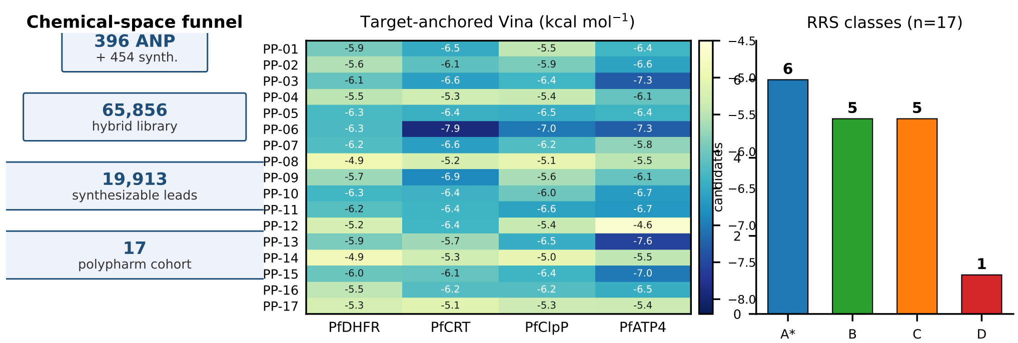
Abstract

Polypharmacology could make antimalarial discovery less vulnerable to single-target resistance, but computational prioritization often conflates multi-target docking, mutation tolerance, and biological activity. Docking is sensitive to receptor and scoring-function choices, so cross-target profiles should be treated as structured hypotheses rather than calibrated measurements.[1,2] We developed an integrated, evidence-bounded workflow that connects an African-natural-product-inspired chemical-space funnel to target-anchored docking and a mutation-aware resistance-resilience score (RRS). A hybrid library of 65 856 molecules, generated from 396 African natural products and 454 synthetic antimalarial compounds, yielded 19 913 computationally prioritized synthesizable leads after variational-autoencoder clustering, multi-parameter optimization, and scaffold expansion. A locked 17-member polypharmacology-oriented cohort was then evaluated by AutoDock Vina against PfDHFR, PfCRT, PfClpP, and PfATP4 using target-specific anchors. All 68 candidate–target pairs passed an automated predeclared geometric gate, with scores ranging from $-7.91\,\mathrm{kcal\,mol^{-1}}$ to $-4.63\,\mathrm{kcal\,mol^{-1}}$; the raw poses remain subject to independent structural review. Separately, corrected per-target RRS was derived from the distinct declared wild-type and mutant docking panel in `docking_mutants.csv` for PfDHFR (N51I, C59R, S108N, I164L) and PfCRT (K76T, K76A), excluding non-binding wild-type targets from the RRS denominator. The 17-member cohort comprised six A*, five B, five C, and one D RRS profiles, with RRS values spanning 68.2 to 111.7. Cross-metric associations were exploratory: PNS–RRS $\rho = -0.559$, ACSI–RRS $\rho = -0.132$, and RRS–wild-type score $\rho = -0.433$; none survived the prespecified Bonferroni threshold. The study yields computational hypotheses about multi-target and mutation-resilient chemotypes, not measured potency, confirmed target engagement, or demonstrated resistance circumvention.



Keywords: African natural products; antimalarial discovery; polypharmacology; resistance resilience; molecular docking; chemical space; PfDHFR; PfCRT

1 Introduction

Antimalarial resistance is a systems problem.[3–5] A compound that depends on one protein–ligand interaction may lose activity after a single parasite mutation,[6] whereas a chemotype that engages several resistance-relevant processes could provide a broader mechanistic starting point.[7,8] This rationale does not imply that every computationally predicted multi-target profile is biologically useful: target breadth, mutation tolerance, physicochemical tractability, and experimental activity are distinct properties that must be measured separately.

African natural products provide a chemically rich but incompletely explored starting point for such studies.[2,9–11] Their ring systems and stereochemical complexity can complement synthetic antimalarial chemistry, while computational expansion can generate tractable analogues around privileged scaffolds. The challenge is to connect this chemical-space exploration to a target-level analysis without allowing the prioritization score to become a surrogate for activity.

We therefore assembled an integrated workflow with three deliberately separated layers. First, a chemical-space funnel combines African natural products and synthetic antimalarials, expands the library, and prioritizes synthesizable candidates. Second, target-anchored docking evaluates a locked polypharmacology-oriented cohort against four proteins with unequal structural evidence. Third, a mutation-aware RRS summarizes changes in docking estimates for declared computational PfDHFR and PfCRT mutant panels. Polypharmacology is read from the four-target docking profile; RRS is read from target-specific mutant comparisons. The two layers are joined by canonical molecular identity for exploratory interpretation, not by assuming that multi-target docking proves mutation resilience.

The central question was: can this separation produce a coherent computational map from chemical-space novelty to multi-target and mutation-resilience hypotheses? We predeclared three safeguards: candidate sets would remain distinct, non-binding wild-type targets would not inflate RRS denominators, and all activity-like language would be restricted to computational predictions.

2 Materials and Methods

2.1 Chemical-space funnel

The upstream library combined 396 African natural products with 454 synthetic antimalarial compounds. Similarity-based expansion and SELFIES perturbation[12,13] were followed by structure validation, deduplication, and PAINS filtering,[14] producing 65 856 unique molecules. A SMILES variational autoencoder[15] was used to obtain latent representations. K-means clustering of the selected 64-dimensional representation yielded 484 centroids. Multi-parameter optimization combined docking-derived scores, DiffDock confidence,[16] QED,[17] an ADMET composite, and a Lipinski penalty.[18] The 76 centroids above the optimization threshold mapped to 53 clusters and expanded to 40 481 molecules; 19 913 molecules remained after synthetic-accessibility and MPO filtering.

Novelty was evaluated with Morgan fingerprints[19] (radius 2, 2 048 bits). In a 5 000-molecule sample, 92.6 % had maximum whole-molecule Tanimoto similarity below 0.4 to the African-NP seed set, while scaffold recovery was 69.3 % (70/101 annotated seed scaffolds). The candidate sets were locked before the target-anchored docking analysis.

2.2 Candidate-set identity

Set A comprised the MPO top-20 candidates, Set B comprised the parent-study MD top-20 candidates, and Set C comprised the 17-member P2 polypharmacology-oriented cohort. The integrated analysis used Set C. Canonical SMILES, candidate identifiers, source hashes, and cohort labels were checked row-wise across the target-panel, P2, and integrated manifests. The four parent-study MD systems (201–PfDHFR, 438–PfATP4, the 164-labelled PfClpR cohort, and 214–PfCRT) were not treated as MD validation of Set C.

2.3 Target-anchored docking

The target panel comprised PfDHFR (7F3Y; methotrexate copy A702), PfCRT (6UKJ; Y01 A501 cavity), PfClpP (2F6I; chain-A Ser252/His223/Asp219 catalytic triad), and PfATP4 (9N10; D451 phosphorylation-loop region and DPPR hinge). The PfClpP identity correction is material: 4GM2 is PfClpR and was excluded from PfClpP claims. Y01 is treated as a membrane/cavity proxy, and the PfATP4 region is mechanism-motivated because no small-molecule inhibitor is co-crystallized.

Ligands were standardized with Open Babel[20] and converted to PDBQT with Meeko.[21] AutoDock Vina was run with fixed target-specific boxes following established docking protocols;[22] scores were interpreted with the usual caveats attached to empirical scoring functions.[23] The geometric gate required a rank-1 centroid inside the box, at least one atom within 10 Å of the target anchor, and at least 90 % of ligand atoms inside the padded box. The gate is a pose-quality and provenance criterion, not evidence of binding. Vina values are reported in kcal mol^{-1} and were not converted to experimental free energies.

2.3.1 Docking protocol validation

The docking protocol was validated on the same target set by three complementary checks. First, re-docking of the co-crystallised reference ligands on the corrected grids reproduced the crystal poses for the five alignable ligands with heavy-atom RMSD < 2.0 Å (100 % success); the PfDHFR reference ligand methotrexate could not be re-docked (RMSD ≈ 30 Å) because the PfDHFR grid is centered on the NADPH-adjacent site rather than the catalytic methotrexate pocket, and is therefore excluded from the re-docking statistic. Second, enrichment against the MMV Malaria Box using the parent-study dual-filter consensus (AutoDock Vina + DiffDock) gave ROC-AUC values of 0.924 to 1.000 across the four targets (0.924 PfDHFR, 0.971 PfCRT, 1.000 PfATP4). Third, an external DEKOIS 2.0 panel for PfDHFR (Vina only, 40 actives / 1 200 property-matched decoys) gave a ROC-AUC of 0.45 (95% CI 0.37 to 0.53), a near-chance result that is reported honestly: it constrains the interpretation of absolute Vina scores for this target and motivates the consensus rescoring and per-target anchors used here, but it does not invalidate the relative within-target rankings or the RRS ratios, which are computed from mutant-to-wild-type score ratios on the same target.

2.4 Resistance-resilience score

RRS was computed separately for each target and candidate. For mutation m and target t,

$$\mathrm{RRS}_{i,m,t} = 100 \times \frac{|S_{\mathrm{Vina},i,m,t}|}{|S_{\mathrm{Vina},i,WT,t}|}. \tag{1}$$

Here S_{Vina} denotes the AutoDock Vina scoring-function output; it is not an experimental free energy. RRS eligibility was evaluated from the separate WT/mutant panel in `docking_mutants.csv`, not from the four-target V5 wild-type matrix. Only targets satisfying the declared genuine-binding wild-type criterion ($|S_{\mathrm{Vina},i,WT,t}| \geq 5\,\mathrm{kcal\,mol^{-1}}$) contributed to a candidate's mean. Non-binding targets were excluded rather than assigned a zero score or used as a denominator. PfDHFR mutations were N51I, C59R, S108N, and I164L; PfCRT mutations were K76T and K76A. RRS classes were assigned using the prespecified P2 rules: A* required the maximum absolute eligible WT score in the mutant panel to be at least $7\,\mathrm{kcal\,mol^{-1}}$ and all available RRS values to be at least 80 %; A required all available values to be at least 80 % without the A* WT-anchor criterion; B required all values to be at least 70 % but not all to be at least 80 %; C required at least one value to be at least 80 % without meeting A/A* or B; and D was assigned when no available mutant RRS reached 80 %. These classes summarize ratios of docking-score magnitudes, not ratios of binding free energies.

2.5 Polypharmacology, PNS, and ACSI

Polypharmacology was treated as a target-profile hypothesis derived from the four-target Vina matrix, not as confirmed simultaneous engagement and not from RRS. No cross-target arithmetic composite was used because Vina scores are not calibrated across different pocket architectures. For the joint prioritization readout in Table 2, a target was counted as favorable when its within-target Vina estimate satisfied the operational threshold $S_{\mathrm{Vina}} \leq -6.0\,\mathrm{kcal\,mol^{-1}}$, chosen a priori as the mid-range of the observed per-target distributions; this is a prioritization device, not an affinity cutoff. PNS was retained as a network-based descriptor and ACSI as a chemical-space descriptor. Their associations with RRS and wild-type Vina score were evaluated with Spearman correlation on the 17-member cohort. Three exploratory correlations were corrected with Bonferroni $\alpha = 0.017$. No association was interpreted as causal, and no non-significant association was interpreted as equivalence.

2.6 Evidence boundary and reproducibility

All raw poses, receptor and ligand files, configurations, candidate manifests, and analysis scripts are archived in the project repository. Automated checks verify file integrity, schema, candidate identity, and row counts. These reproducibility controls do not substitute for structural or experimental validation; the manuscript therefore reports the computational evidence together with its uncertainty and scope.

3 Results

3.1 Chemical-space expansion preserved scaffolds while diversifying molecules

The hybrid library contained 65 856 unique structures. Whole-molecule novelty and scaffold recovery were simultaneously high: 92.6 % of the evaluated sample was below the whole-molecule Tanimoto threshold of 0.4, whereas 70 of 101 annotated seed scaffolds were recovered. Scaffold-only similarity exceeded whole-molecule similarity (mean 0.379 versus 0.206), consistent with preservation of ring systems and diversification of peripheral substituents. The funnel produced 19 913 computationally prioritized molecules predicted to be synthesizable; this is a prioritization output, not a synthesis result.

3.2 The locked cohort spans four computational target profiles

Set C contained 17 candidates selected for a polypharmacology-oriented analysis. Target-anchored docking generated 68/68 pairs that passed the automated geometric gate. Score ranges were $-6.35\,\mathrm{kcal\,mol^{-1}}$ to $-4.86\,\mathrm{kcal\,mol^{-1}}$ for PfDHFR, $-7.91\,\mathrm{kcal\,mol^{-1}}$ to $-5.12\,\mathrm{kcal\,mol^{-1}}$ for PfCRT, $-7.03\,\mathrm{kcal\,mol^{-1}}$ to $-5.05\,\mathrm{kcal\,mol^{-1}}$ for PfClpP, and $-7.57\,\mathrm{kcal\,mol^{-1}}$ to $-4.63\,\mathrm{kcal\,mol^{-1}}$ for PfATP4. This 68-pair result is raw computational evidence from an automated pose-quality screen; the gate is not independent structural review and does not constitute biological validation. The most favorable within-target estimates included PP-06 in the PfCRT cavity (−7.91) and PP-13 in the PfATP4 machinery region (−7.57). These are target-specific, within-panel observations rather than calibrated potency rankings; no cross-target arithmetic composite was used.

3.3 Per-target RRS separates mutation tolerance from target breadth

Corrected RRS analysis classified the 17-member cohort as six A*, five B, five C, and one D profiles. RRS values ranged from 68.2 to 111.7. Five candidates were classified on PfCRT alone because their PfDHFR WT estimates in the separate mutant docking panel did not meet the genuine-binding threshold; this statement does not refer to the distinct V5 four-target wild-type matrix. This target-specific missingness is scientifically informative: it prevents a near-zero wild-type denominator from producing an artificial resistance-resilience score.

The previously described PP-11 PfDHFR C59R knockout pattern was not retained as a design rule. PP-11's PfDHFR WT estimate in the separate mutant panel was non-binding under the declared threshold, whereas its PfCRT mutant profile was evaluated on the PfCRT denominator; the four-target V5 matrix is not used to define this RRS eligibility decision. The corrected analysis therefore supports a target-specific rather than a mixed-target interpretation of resilience. The complete per-mutant profile is listed in Table 1 and visualized in Figure 1; cross-metric relationships are displayed in Figure 2. These derived figures and tables are reproducible computational outputs; they should not be interpreted as independent structural validation or experimental evidence.

Table 1: Per-target resistance-resilience scores (RRS, Equation 1) for the 17-member cohort. Values are percentages of the corresponding wild-type Vina score magnitude; — marks wild-type non-binders ($|S_{\text{Vina},i,WT,t}| < 5\,\text{kcal}\,\text{mol}^{-1}$), excluded from the denominator. RRS_{mean} is the candidate mean over all eligible target–mutant ratios. Classes are the operational categories defined in Table S2, not biological resistance phenotypes.

Candidate	Class	$\text{RRS}_{\text{PfDHFR}}$ (%)				$\text{RRS}_{\text{PfCRT}}$ (%)		RRS_{mean}
		N51I	C59R	S108N	I164L	K76T	K76A	(%)
PP-01	A*	86.4	85.5	87.6	83.9	92.5	90.9	87.8
PP-02	A*	—	—	—	—	87.9	94.6	91.3
PP-03	C	68.1	68.3	65.8	69.0	80.0	81.4	72.1
PP-04	C	68.8	66.9	68.8	73.5	105.3	103.6	81.1
PP-05	B	—	—	—	—	77.1	79.3	78.2
PP-06	A*	—	—	—	—	90.6	94.1	92.4
PP-07	B	71.8	71.8	72.4	72.2	84.2	77.6	75.0
PP-08	C	62.1	62.1	67.2	66.3	83.9	83.1	70.8
PP-09	B	70.6	73.0	75.8	76.5	75.8	73.5	74.2
PP-10	B	76.0	76.4	75.7	75.8	89.4	89.6	80.5
PP-11	A*	—	—	—	—	82.7	82.5	82.6
PP-12	C	75.2	66.1	62.9	72.5	70.4	87.2	72.4
PP-13	A*	—	—	—	—	110.2	99.4	104.8
PP-14	D	62.2	65.5	68.0	64.9	78.8	77.7	69.5
PP-15	A*	123.2	112.7	125.9	131.8	86.2	90.1	111.7
PP-16	B	72.9	74.7	73.7	75.9	80.1	80.6	76.3
PP-17	C	59.3	61.2	59.5	58.9	76.8	93.4	68.2

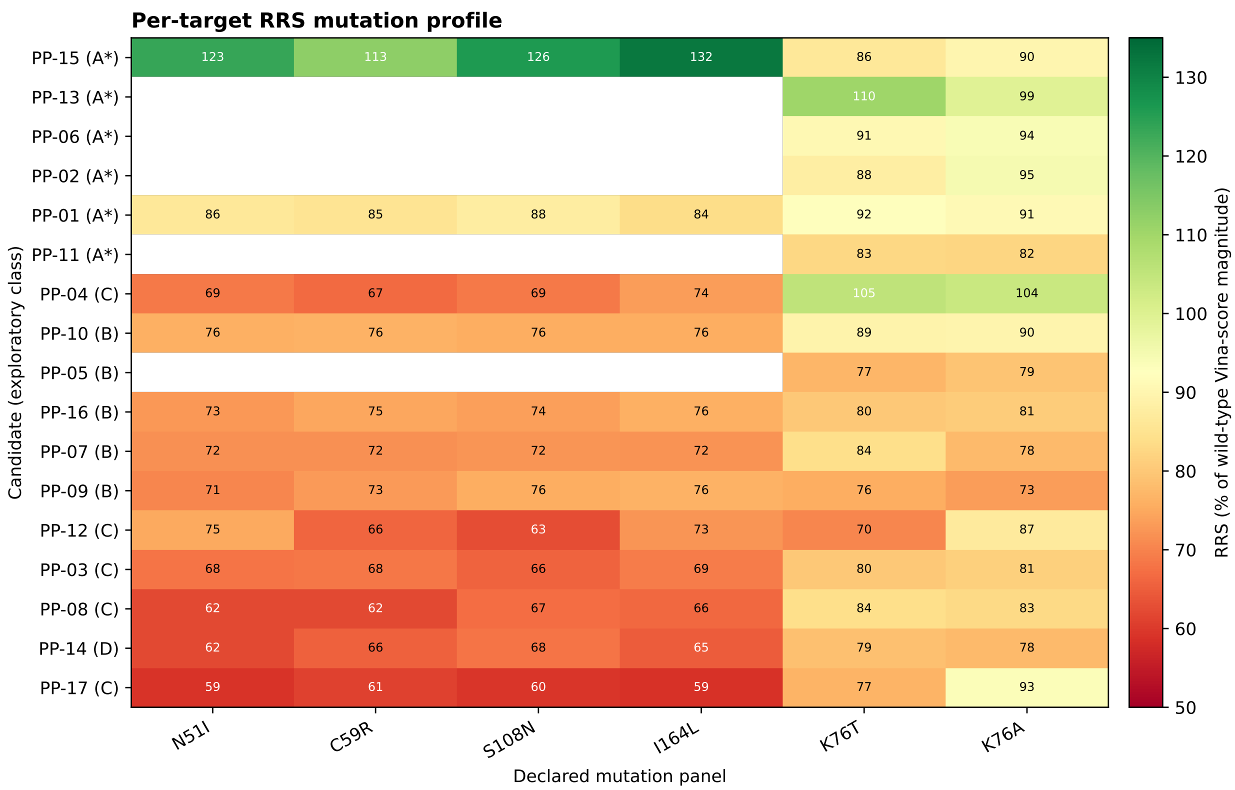
3.4 Combining target breadth with mutation tolerance ranks the candidates

To make the joint polypharmacology–RRS readout explicit, each candidate was summarized by the number of targets with a favorable within-target Vina estimate (defined operationally as $S_{\mathrm{Vina}} \leq -6.0\,\mathrm{kcal\,mol^{-1}}$, the mid-range of the observed per-target distributions) alongside its RRS class and mean. The resulting ordering, in Table 2, is an evidence-aggregation device, not a calibrated activity ranking.

Table 2: Combined polypharmacology–RRS prioritization of the 17-member cohort. N_{fav} counts targets with a favorable within-target Vina estimate ($S_{\mathrm{Vina}} \leq -6.0\,\mathrm{kcal\,mol^{-1}}$); favorable targets are listed explicitly. Candidates are sorted by N_{fav} and then by $\mathrm{RRS_{mean}}$. The threshold is an operational prioritization device, not an experimental affinity cutoff, and the classes are the operational RRS categories of Table S2.

Candidate	RRS class	$\mathrm{RRS_{mean}}$ (%)	N_{fav}	Favorable targets
PP-15	A*	111.7	4	PfDHFR + PfCRT + PfClpP + PfATP4
PP-06	A*	92.4	4	PfDHFR + PfCRT + PfClpP + PfATP4
PP-11	A*	82.6	4	PfDHFR + PfCRT + PfClpP + PfATP4
PP-10	B	80.5	4	PfDHFR + PfCRT + PfClpP + PfATP4
PP-05	B	78.2	4	PfDHFR + PfCRT + PfClpP + PfATP4
PP-03	C	72.1	4	PfDHFR + PfCRT + PfClpP + PfATP4
PP-16	B	76.3	3	PfCRT + PfClpP + PfATP4
PP-07	B	75.0	3	PfDHFR + PfCRT + PfClpP
PP-13	A*	104.8	2	PfClpP + PfATP4
PP-02	A*	91.3	2	PfCRT + PfATP4
PP-01	A*	87.8	2	PfCRT + PfATP4
PP-09	B	74.2	2	PfCRT + PfATP4
PP-04	C	81.1	1	PfATP4
PP-12	C	72.4	1	PfCRT
PP-08	C	70.8	0	—
PP-14	D	69.5	0	—
PP-17	C	68.2	0	—

Three candidates dominate the joint readout: PP-15, PP-06, and PP-11 are the only members that combine a favorable estimate on all four targets with an A* RRS classification. They differ in the resistance channel: PP-15 ($\mathrm{RRS_{mean}}$ 111.7) and PP-06 (92.4) are resilient on the PfCRT mutant panel, while PP-11 (82.6) is PfCRT-classified with PfDHFR non-binding in the mutant panel. By contrast, PP-13—previously highlighted on its Pareto-front position—combines the second-highest RRS mean (104.8) with favorable estimates on only two targets, illustrating that mutation tolerance and target breadth are genuinely independent axes: joint ranking, not either score alone, identifies the most economical multi-target, mutation-resilient hypotheses.



Exploratory source-panel ratios; blank cells denote excluded non-binding wild-type targets.

Figure 1: Exploratory per-target RRS mutation profiles for the 17-member source-panel cohort. Values are ratios of Vina-score magnitudes relative to the declared wild-type denominator; blank cells indicate excluded non-binding wild-type targets. The classes are computational operational categories, not biological resistance phenotypes.

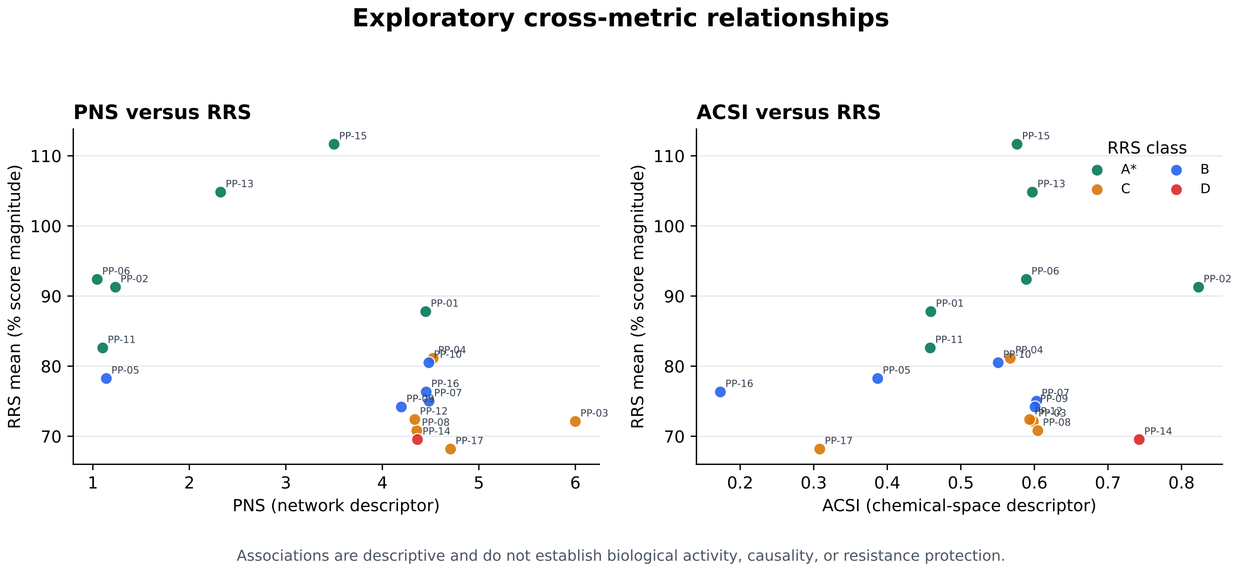


Figure 2: Exploratory relationships between RRS and PNS or ACSI. Points are colored by the operational RRS class. These associations are descriptive source-panel analyses and do not establish causality, biological activity, or resistance protection.

3.5 Cross-metric associations were exploratory and mostly null

On the 17-member cohort, PNS and RRS showed a negative trend ($\rho = -0.559$), but the nominal result did not survive Bonferroni correction. ACSI and RRS were weakly associated ($\rho = -0.132$), and RRS versus wild-type Vina score showed $\rho = -0.433$; neither was significant under the prespecified threshold. PNS versus wild-type Vina score was positive ($\rho = 0.389$), illustrating that network and docking descriptors need not be interchangeable. These results do not establish a general relationship between natural-product character, target breadth, and mutation tolerance.

4 Discussion

The principal contribution of this study is an evidence architecture rather than a claim of confirmed multi-target activity. In particular, the workflow makes target-specific evidence classes explicit and prevents an uncalibrated cross-target score average from disguising differences in pocket quality. The chemical-space funnel provides structurally diverse, computationally tractable hypotheses; target-anchored docking provides a standardized pose layer; and per-target RRS provides a mutation-aware comparison that avoids target mixing. Keeping these layers separate makes the resulting hypotheses more falsifiable.

The four-target profile and the RRS profile answer different questions. A candidate can have favorable docking estimates across several targets without being resilient to a specific mutation. Conversely, a candidate can show high RRS on one target while lacking evidence for a second target. Because raw Vina scores are not calibrated across unlike receptors, target breadth is represented by the pattern of target-specific evidence, not by averaging scores across targets. The use of a 17-member locked cohort and canonical-SMILES joins prevents these distinctions from being hidden by row order or by a single composite score.

The null and corrected findings are important. The absence of a Bonferroni-significant cross-metric association means that chemical-space relatedness and network position cannot be used as substitutes for target-specific mutation analysis. The retraction of the PP-11 C59R interpretation further demonstrates why non-binding wild-type baselines must be excluded before calculating ratios. In this setting, a narrower and more conditional RRS claim is stronger than an apparently larger pooled-target effect.

Reading the two layers together is more informative than either alone, and the joint ranking in Table 2 makes that readout explicit. PP-15, PP-06, and PP-11 are the only candidates with a favorable Vina estimate on all four targets together with an A* RRS classification; PP-15 adds the highest cohort RRS mean (111.7%), PP-06 the most favorable PfCRT-cavity estimate (-7.91 kcal mol^{-1}) with an A* PfCRT profile (92.4%), and PP-11 an A* PfCRT profile (82.6%) despite PfDHFR non-binding in the mutant panel. These three candidates therefore represent the workflow's most economical combined multi-target and mutation-resilience hypotheses and its most defensible first choices for biochemical, mutant-panel, and molecular-dynamics follow-up. PP-13, despite its second-highest RRS mean (104.8%), is favorable on only two targets—an instructive counterexample showing that mutation tolerance and target breadth are independent axes. The remaining A* assignments are classified on the PfCRT channel only (PP-02, PP-06, PP-11, PP-13) or on both PfDHFR and PfCRT channels (PP-01, PP-15), with PP-15 additionally favorable on all four Vina targets; none of these operational assignments should be read as confirmed activity.

The study has substantial limitations. Docking scores are approximate scoring-function outputs and do not establish affinity, kinetics, or cellular activity. PfCRT is represented by a proxy cavity, and PfATP4 by conserved catalytic machinery without a co-crystallized inhibitor. RRS covers PfDHFR and PfCRT but not PfClpP or PfATP4, and the cohort is small and purposefully enriched for predicted polypharmacology. The parent MD systems are separate from Set C and cannot validate the Set C RRS profile. No new IC_{50}/EC_{50} measurements were available. Finally, independent structural assessment remains necessary before the raw target panel can support stronger structural conclusions. The next experiments should combine biochemical or cellular potency measurements, resistance-mutant assays, orthogonal binding methods, and candidate-specific molecular dynamics.

5 Conclusion

An African-natural-product-inspired chemical-space funnel was connected to a four-target, target-anchored docking panel and a corrected per-target RRS analysis. The workflow identified a 17-member cohort with reproducible computational pose profiles and heterogeneous mutation-resilience classes, while retaining null findings and target-specific missingness. Its value is hypothesis generation: it proposes which chemotypes and target–mutation combinations should be tested next, without claiming that docking, RRS, or polypharmacology estimates are experimental evidence.

Supporting Information

The Supporting Information contains the candidate manifest, target-anchor evidence, complete 17-by-4 docking matrix, RRS definitions and table, PNS/ACSI values, cross-metric statistics, provenance fields, and reproducibility instructions.

Author Contributions

M.V.S.T. and S.G.N.E. designed the integrated study. M.V.S.T. and J.-P.T.N. developed the chemical-space and docking analyses. P.S. and W.F.M. contributed to target biology and resistance interpretation. All authors reviewed and approved the manuscript scope.

Notes

The authors declare no competing financial interest. This is a computational, hypothesis-generating study; no experimental potency is claimed.

Data Availability

Data and code are available at https://github.com/NanaEngo/Malaria_codesV2. Integrated manifests and derived figures are in `Project1_Chem_space_antimalarial_V6_CorrectedGrid/`; source RRS data are in `Project2_Polypharmacology_MD_ValidationV2607/results/`; raw Vina data are in `Project1_Chem_space_antimalarial_V5_CorrectedGrid/results/`.

Use of Artificial Intelligence

During the preparation of this work the authors used AI assistants to support code development and data analysis scripting. The authors reviewed and edited all AI-assisted output and take full responsibility for the content of this article. No generative AI tool is listed as an author. All computational results, statistical analyses, and quantitative claims were generated and verified by the authors.

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